THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

UBDA
UNIVERSITY RESEARCH FACILITY IN BIG DATA ANALYTICS
大數據分析中心實驗室

UBDA Platform

Example for Tensorflow

User Guide

Version 1.0

16 July 2018

Revision History

Version	Date	Prepared By	Summary of Changes
1.0	Jul 16, 2018		Initial release

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1. Introduction

This document is shown a Tensorflow examplerunning on the UBDA platform.

Note: User should first register a user account through UBDA website at: https://www.polyu.edu.hk/pfs/index.php/177729 to access the UBDA Platform

2. Perform the test

2.1 Login to ubdaplatform.polyu.edu.hk via SSH

2.2 Create the testing directory

```
$ mkdir -p $HOME/tensorflow
$ cd $HOME/tensorflow
```

2.3 Prepare the tensorflow program (Filename: cifar10_multi_gpu_train.py)

```
$  cp -pr /ubda/apps/examples/tensorflow/models .
$  cp -pr /ubda/apps/examples/tensorflow/data .
```

2.4 Prepare the job scripte file.

tensorflow.pbs

```
#!/bin/sh
#PBS -N tensorflow-test
#PBS -l nodes=1:ppn=1 -W x=GRES:gpu@4
#PBS -l walltime=12:00:00
#PBS -q qgpu01
#PBS -V
#PBS -j oe
#PBS -S /bin/bash

module load cuda-9.0
module load python-3.6.6

###############################################################
##
NP=`cat $PBS_NODEFILE | wc -l`
NN=`cat $PBS_NODEFILE | sort | uniq | tee /tmp/nodes.$$ | wc -
l`
GPU=`cat $PBS_GPUFILE | wc -l`
cat $PBS_NODEFILE > /tmp/nodefile.$$

echo "process will start at : "
date
echo "++++++++++++++++++++++++++++++++++++++++"
cd $PBS_O_WORKDIR

export PATH=/ubda/apps/python/python-3.6.6/bin:$PATH

python3
$HOME/tensorflow/models/tutorials/image/cifar10/cifar10_multi_
gpu_train.py --num_gpus=4 > result.out
rm -rf /tmp/cifar10_data
```

```
rm -rf /tmp/cifar10_train
echo "++++++++++++++++++++++++++++++++++++++++"
echo "processs will sleep 30 seconds"
sleep 30
echo "process end at : "
date
rm -f /tmp/nodefile.$$
rm -f /tmp/nodes.$$


module unload cuda-9.0
module unload python-3.6.6
```

Please remind to change the values for your application.

```
#PBS -N tensorflow-test {your job name}
#PBS -l nodes=1:ppn=1 -W x=GRES:gpu@4
{your requested resource;nodes and processors per node; -W
x=GRES:gpu@4 your gpu number }
#PBS -q qgpu01{the job queue}
--num_gpus=4 {your gpu number}
```

Example files can be found at:
/ubda/apps/examples/tensorflow

model
data
tensorflow.pbs

3 Job submission

3.1 Submit the script (***tensorflow.pbs)*** to job queue.
A job ID number will be returned.

```
$ qsub tensorflow.pbs
1204.ubda-mgt01
```

3.2 Enquiry the submitted job status.

```
$ qstat -na

ubda-mgt01:
                                                                    Req'd       Req'd       Elap
Job ID                    Username    Queue    Jobname          SessID  NDS   TSK   Memory      Time     S   Time
------------------------ ----------- -------- ---------------- ------ ----- ------ --------- --------- - ---------
1204.ubda-mgt01            ubdademo9   qgpu01   tensorflow-test       --      1      1        --   12:00:00 Q
```

Job status field name	Explanation	Example
JOB ID	Unique Job id.	1204.ubda-mgt01
Job name	Name for the job allocation	tensorflow-test
Queue name	Name of the job queue that the job has assigned.	gpu01
Username	Your NetID	ubdademo9
S	Job current status. Q = queued R = Job is executing C = Job was completed	Queued
ELAP TIME	Time for the job executed	
Req’d Time	The maximum execution time	12:00:00 (1 day)
NDS	Total number of nodes assigned	1
TSK	Total number of cores assigned	1

4 How to check the result

4.1 Check for any error messages for JobID

```
$ more tensorflow-test.e1204
Host key verification failed.(know error message can be ignored)

$ more tensorflow-test.o1204
process will start at :
Tue Jul  3 16:41:59 HKT 2018
++++++++++++++++++++++++++++++++++++++++
2018-07-03 16:42:16.083478: I
tensorflow/core/platform/cpu_feature_guard.cc:140] Your CPU
supports instructions that this TensorFlow binary was not compiled
to use: AVX2 AVX512F FMA
2018-07-03 16:42:16.387654: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:1356] Found
device 0 with properties:
name: Tesla P100-PCIE-16GB major: 6 minor: 0 memoryClockRate(GHz):
1.3285
pciBusID: 0000:88:00.0
totalMemory: 15.90GiB freeMemory: 15.61GiB
2018-07-03 16:42:16.564489: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:1356] Found
device 1 with properties:
name: Tesla P100-PCIE-16GB major: 6 minor: 0 memoryClockRate(GHz):
1.3285
pciBusID: 0000:8d:00.0
totalMemory: 15.90GiB freeMemory: 15.61GiB
2018-07-03 16:42:16.747348: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:1356] Found
device 2 with properties:
name: Tesla P100-PCIE-16GB major: 6 minor: 0 memoryClockRate(GHz):
1.3285
pciBusID: 0000:b3:00.0
totalMemory: 15.90GiB freeMemory: 15.61GiB
2018-07-03 16:42:16.935390: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:1356] Found
device 3 with properties:
name: Tesla P100-PCIE-16GB major: 6 minor: 0 memoryClockRate(GHz):
1.3285
pciBusID: 0000:b6:00.0
totalMemory: 15.90GiB freeMemory: 15.61GiB
2018-07-03 16:42:16.945848: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:1435] Adding
visible gpu devices: 0, 1, 2, 3
2018-07-03 16:42:18.348596: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:923] Device
interconnect StreamExecutor with strength 1 edge matrix:
2018-07-03 16:42:18.348810: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:929]      0 1 2 3
2018-07-03 16:42:18.348960: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:942] 0:   N Y Y Y
```

```
2018-07-03 16:42:18.349073: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:942] 1:   Y N Y Y
2018-07-03 16:42:18.349192: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:942] 2:   Y Y N Y
2018-07-03 16:42:18.349306: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:942] 3:   Y Y Y N
2018-07-03 16:42:18.350574: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:1053] Created
TensorFlow device (/job:localhost/replica:0/task:0/device:GPU:0
with 15131 MB memory) -> physical GPU (device: 0, name: Tesla
P100-PCIE-16GB, pci bus id: 0000:88:00.0, compute capability: 6.0)
2018-07-03 16:42:18.497018: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:1053] Created
TensorFlow device (/job:localhost/replica:0/task:0/device:GPU:1
with 15131 MB memory) -> physical GPU (device: 1, name: Tesla
P100-PCIE-16GB, pci bus id: 0000:8d:00.0, compute capability: 6.0)
2018-07-03 16:42:18.643130: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:1053] Created
TensorFlow device (/job:localhost/replica:0/task:0/device:GPU:2
with 15131 MB memory) -> physical GPU (device: 2, name: Tesla
P100-PCIE-16GB, pci bus id: 0000:b3:00.0, compute capability: 6.0)
2018-07-03 16:42:18.789231: I
tensorflow/core/common_runtime/gpu/gpu_device.cc:1053] Created
TensorFlow device (/job:localhost/replica:0/task:0/device:GPU:3
with 15131 MB memory) -> physical GPU (device: 3, name: Tesla
P100-PCIE-16GB, pci bus id: 0000:b6:00.0, compute capability: 6.0)
Host key verification failed.
cgdelete: cannot remove group 'ubda-g002.1107.ubda-mgt01': Device
or resource busy
```

4.2 Check for result

```
$ head -50 result.out
Filling queue with 20000 CIFAR images before starting to train.
This will take a few minutes.
2018-06-28 15:53:55.106422: step 0, loss = 4.68 (133.0
examples/sec; 0.962 sec/batch)
2018-06-28 15:53:55.815010: step 10, loss = 4.60 (30070.9
examples/sec; 0.004 sec/batch)
2018-06-28 15:53:55.995739: step 20, loss = 4.55 (26906.4
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:56.174933: step 30, loss = 4.43 (24494.0
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:56.350941: step 40, loss = 4.31 (31085.1
examples/sec; 0.004 sec/batch)
2018-06-28 15:53:56.527439: step 50, loss = 4.26 (31282.5
examples/sec; 0.004 sec/batch)
2018-06-28 15:53:56.702668: step 60, loss = 4.31 (30824.1
examples/sec; 0.004 sec/batch)
2018-06-28 15:53:56.883860: step 70, loss = 4.10 (25653.8
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:57.085617: step 80, loss = 4.17 (21655.2
examples/sec; 0.006 sec/batch)
```

```
2018-06-28 15:53:57.357855: step 90, loss = 4.01 (20844.1
examples/sec; 0.006 sec/batch)
2018-06-28 15:53:57.594618: step 100, loss = 3.95 (22861.9
examples/sec; 0.006 sec/batch)
2018-06-28 15:53:57.950736: step 110, loss = 4.00 (26238.4
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:58.156082: step 120, loss = 4.16 (26643.7
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:58.352440: step 130, loss = 3.90 (24242.9
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:58.529255: step 140, loss = 3.86 (31454.4
examples/sec; 0.004 sec/batch)
2018-06-28 15:53:58.709189: step 150, loss = 3.90 (27085.6
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:58.887643: step 160, loss = 3.90 (29978.6
examples/sec; 0.004 sec/batch)
2018-06-28 15:53:59.072527: step 170, loss = 3.87 (26806.4
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:59.245500: step 180, loss = 3.63 (25268.7
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:59.430830: step 190, loss = 3.73 (26966.2
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:59.629899: step 200, loss = 3.67 (27751.4
examples/sec; 0.005 sec/batch)
2018-06-28 15:53:59.961751: step 210, loss = 3.62 (26263.1
examples/sec; 0.005 sec/batch)
2018-06-28 15:54:00.143679: step 220, loss = 3.92 (28466.5
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:00.325511: step 230, loss = 3.61 (29672.9
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:00.504568: step 240, loss = 3.53 (21249.0
examples/sec; 0.006 sec/batch)
2018-06-28 15:54:00.676512: step 250, loss = 3.85 (28681.3
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:00.855951: step 260, loss = 3.73 (28175.2
examples/sec; 0.005 sec/batch)
2018-06-28 15:54:01.027358: step 270, loss = 3.52 (29159.9
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:01.208308: step 280, loss = 3.58 (31082.4
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:01.382450: step 290, loss = 3.58 (29934.7
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:01.563893: step 300, loss = 3.76 (24545.5
examples/sec; 0.005 sec/batch)
2018-06-28 15:54:01.875053: step 310, loss = 3.39 (24414.6
examples/sec; 0.005 sec/batch)
2018-06-28 15:54:02.052311: step 320, loss = 3.55 (30062.5
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:02.236507: step 330, loss = 3.51 (28243.4
examples/sec; 0.005 sec/batch)
2018-06-28 15:54:02.420398: step 340, loss = 3.31 (27049.8
examples/sec; 0.005 sec/batch)
2018-06-28 15:54:02.595836: step 350, loss = 3.34 (32340.6
examples/sec; 0.004 sec/batch)
```

```
2018-06-28 15:54:02.775875: step 360, loss = 3.53 (28796.7
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:02.960746: step 370, loss = 3.43 (28689.0
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:03.140107: step 380, loss = 3.49 (32167.2
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:03.314249: step 390, loss = 3.35 (29903.0
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:03.492490: step 400, loss = 3.20 (29260.1
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:03.800246: step 410, loss = 3.33 (30596.6
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:03.977136: step 420, loss = 3.38 (32250.8
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:04.148625: step 430, loss = 3.17 (31236.6
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:04.320711: step 440, loss = 3.11 (29627.8
examples/sec; 0.004 sec/batch)
2018-06-28 15:54:04.492442: step 450, loss = 3.31 (28091.1
examples/sec; 0.005 sec/batch)
2018-06-28 15:54:04.677241: step 460, loss = 3.17 (24712.7
examples/sec; 0.005 sec/batch)
2018-06-28 15:54:04.889363: step 470, loss = 3.16 (26625.5
examples/sec; 0.005 sec/batch)
2018-06-28 15:54:05.090059: step 480, loss = 3.09 (28804.0
examples/sec; 0.004 sec/batch)
```

5 Useful Reference

- Tutorials of tensorflow
  URL: https://www.tensorflow.org/tutorials/

- Command reference for qstat
  URL: http://docs.adaptivecomputing.com/torque/6-0-0/help.htm#topics/torque/commands/qstat.htm?Highlight=qstat